Music Genre Classification Using Machine Learning

Executive Summary
This project proposes the development of a simple, scalable machine learning model capable of classifying music into predefined genres based on audio input. The solution leverages lightweight audio feature extraction techniques and a traditional supervised learning algorithm to produce accurate genre predictions with minimal complexity.

Objectives
- Develop a machine learning model that accepts audio input and predicts a genre label
- Utilize simple feature extraction techniques
- Implement a lightweight classification algorithm
- Deliver a working prototype with performance metrics
- Establish a modular pipeline for future use

In Scope:
- Dataset selection (5–10 genres)
- Audio preprocessing and feature extraction
- Model training and evaluation
- Prototype development

Out of Scope:
- Real-time processing
- Deep learning models
- Production deployment

Technical Approach
Pipeline: Audio Input → Feature Extraction → Feature Vector → ML Model → Genre Prediction

Tools:
- Python
- Librosa
- Pandas / NumPy
- Scikit-learn
- Kaggle
- Hugging Face

Model:
- Random Forest (primary)
- Optional: k-NN, Logistic Regression, SVM

Deliverables
- Feature extraction pipeline
- Trained model

- Prototype script
- Technical documentation

Timeline
Estimated duration: 2–3 weeks
- Setup: 2–3 days
- Feature engineering: 3–5 days
- Model training: 3–5 days
- Evaluation: 2–3 days
- Documentation: 2–3 days

Risks
- Small dataset → Mitigation: balanced data
- Genre ambiguity → Limit genres
- Overfitting → Cross-validation

Success Criteria
- Model predicts genres correctly
- Achieves ~60–80% accuracy
- Pipeline is modular and reusable

Conclusion
This project delivers a low-complexity machine learning solution for music genre classification, providing a strong foundation for future enhancements.